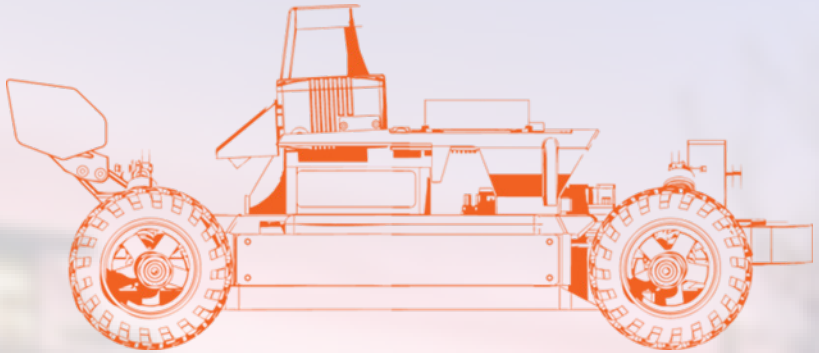


LAB-01

Developer Onramp



Chinmay Vilas Samak

Ph.D. Candidate

ARMLab, CU-ICAR

Email: csamak@clemson.edu



Tanmay Vilas Samak

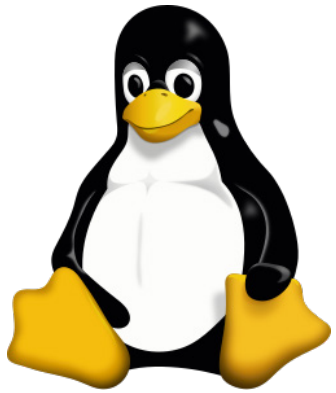
Ph.D. Candidate

ARMLab, CU-ICAR

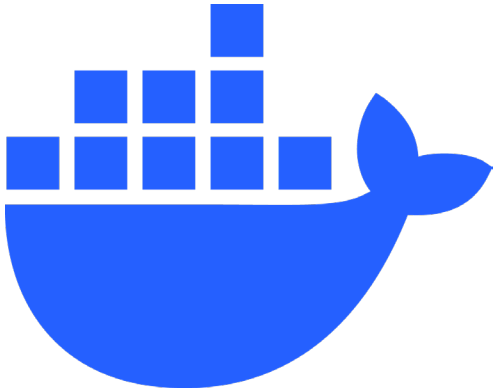
Email: tsamak@clemson.edu

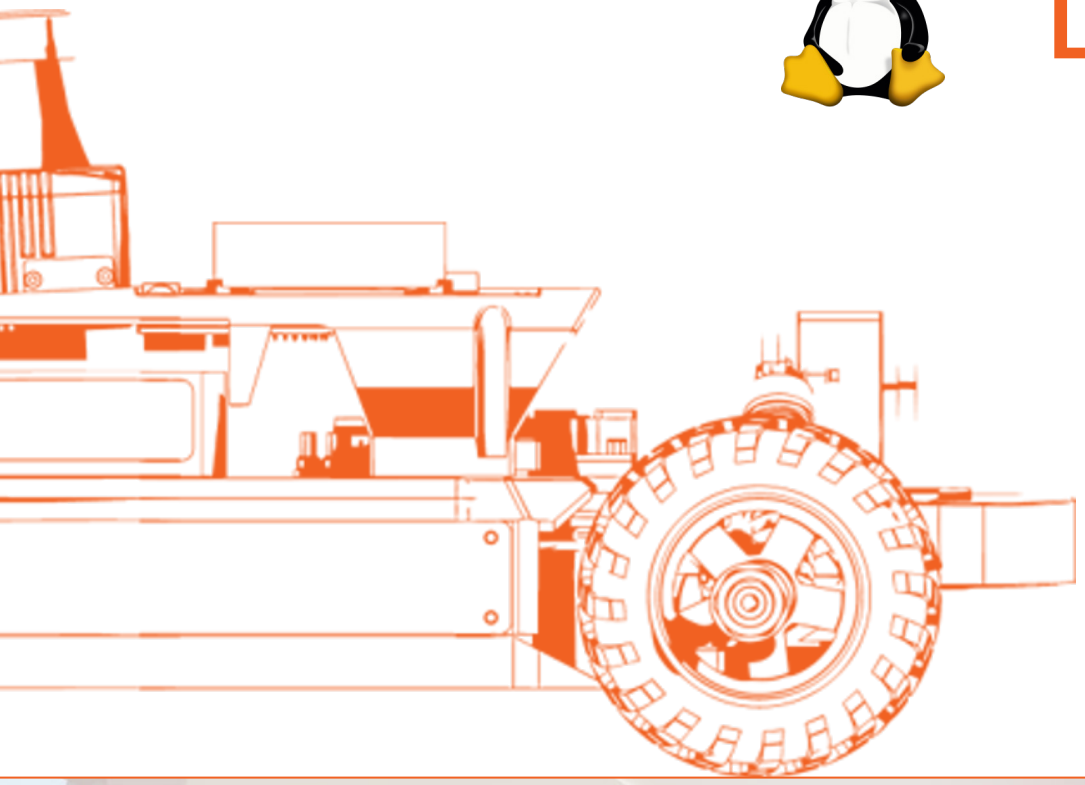
Lab Outline and Expected Outcomes

- **Development**
  - Linux (Ubuntu 22.04)
  - Windows Subsystem for Linux (WSL 2)
- **Version Control**
  - Git
  - GitHub
- **Containerization**
  - Docker
  - Docker Hub
- **GOAL:** Develop, collaborate, package & deploy seamlessly









Linux 101

Linux-101.md

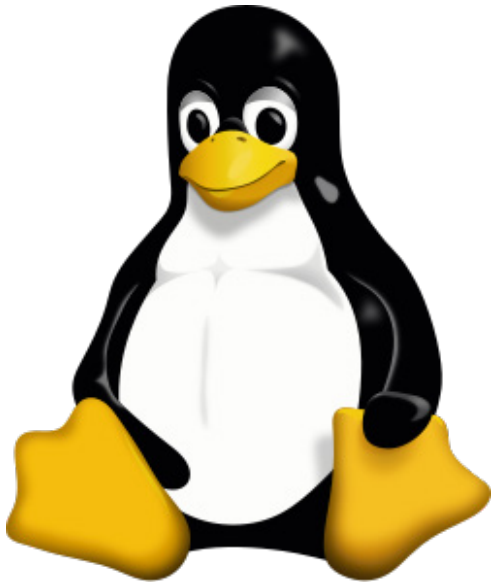
Linux 101

Linux makes a developer's life easy!

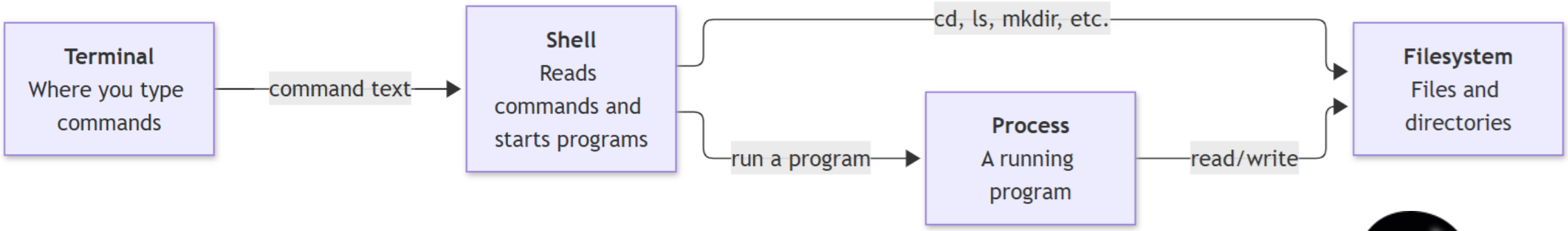
Course context: Autonomous vehicle projects often run on Linux because robotics tools, simulators, build systems, and embedded computing workflows are commonly developed for Linux.

Learning goals:

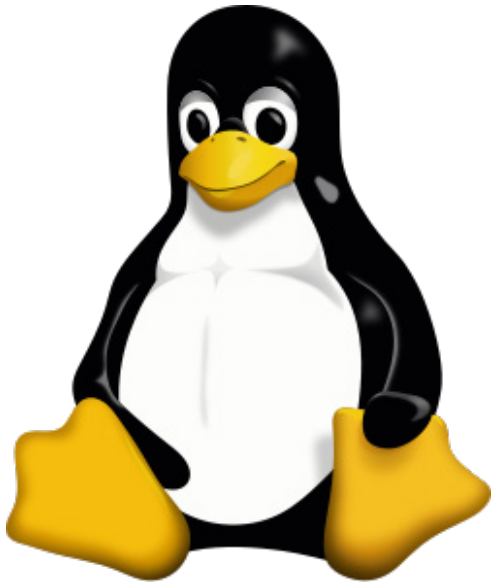
- Navigate the filesystem
- Create, inspect, copy, move, and remove files
- Search for files and text
- Understand paths, permissions, commands, and processes
- Manage packages with apt
- Use command history, redirection, and pipes
- Find help when a command is unfamiliar



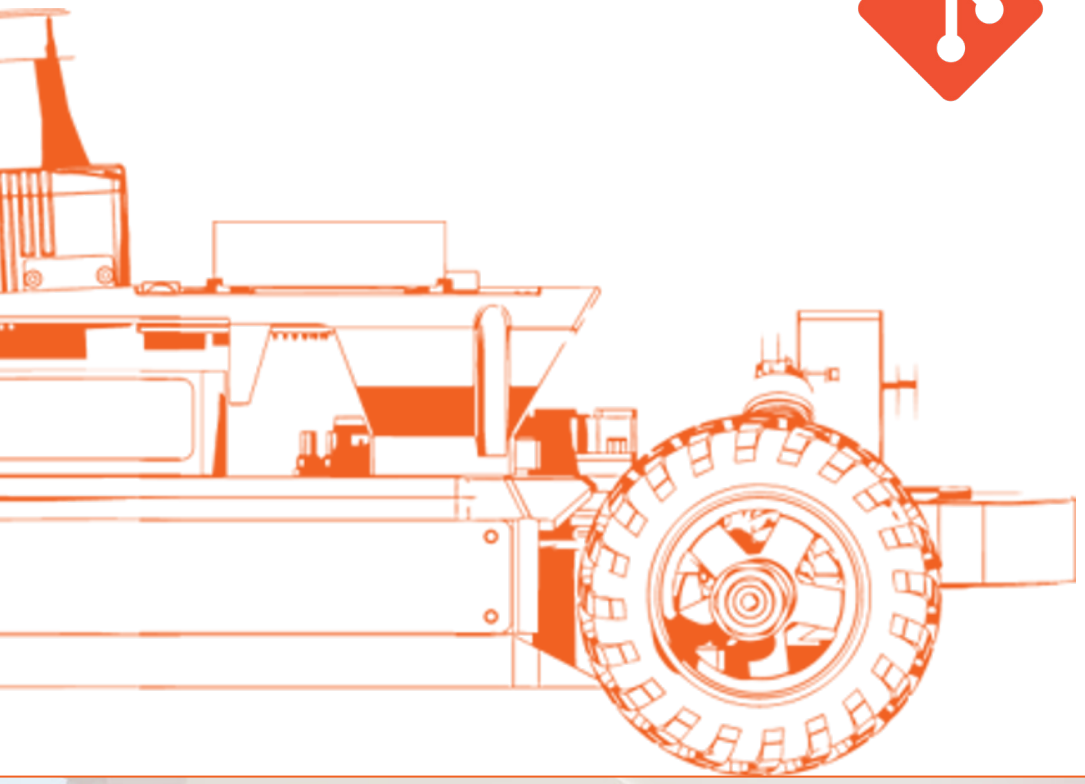
Linux 101



- The **terminal** is the window where you type commands and see output
- The **shell** is the program that reads your commands; Bash is a common shell on Ubuntu
- The **filesystem** is the tree of directories and files beginning at /
- A **process** is a program that is currently running
- Most **commands** either inspect or change the filesystem, start a process, or connect these ideas together with pipes and redirection



Git 101



Git-101.md

Git 101

Git turns edits into traceable history!

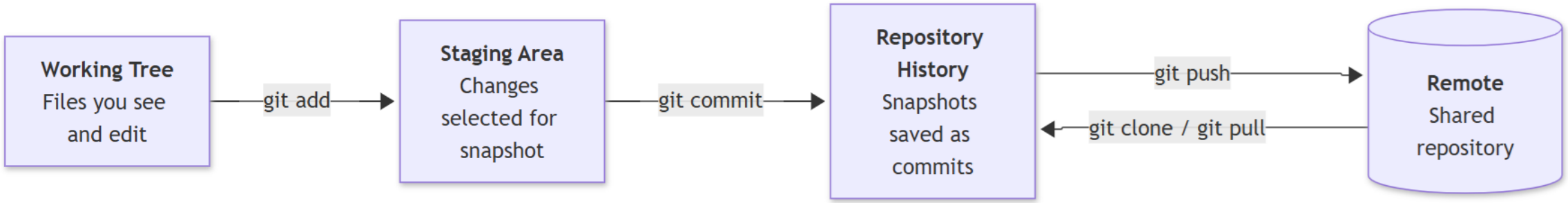
Course context: Autonomous racing teams use Git to track changes to the autonomy stack, simulation configurations, and documentation. GitHub provides a shared platform for reviewing, testing, and integrating the project efficiently.

Learning goals:

- Create or clone a Git repository
- Understand working-tree, staging-area, and commit states
- Make focused commits
- Inspect history and changes
- Work with branches
- Connect a local repository to GitHub
- Pull and push changes
- Handle merge conflicts safely



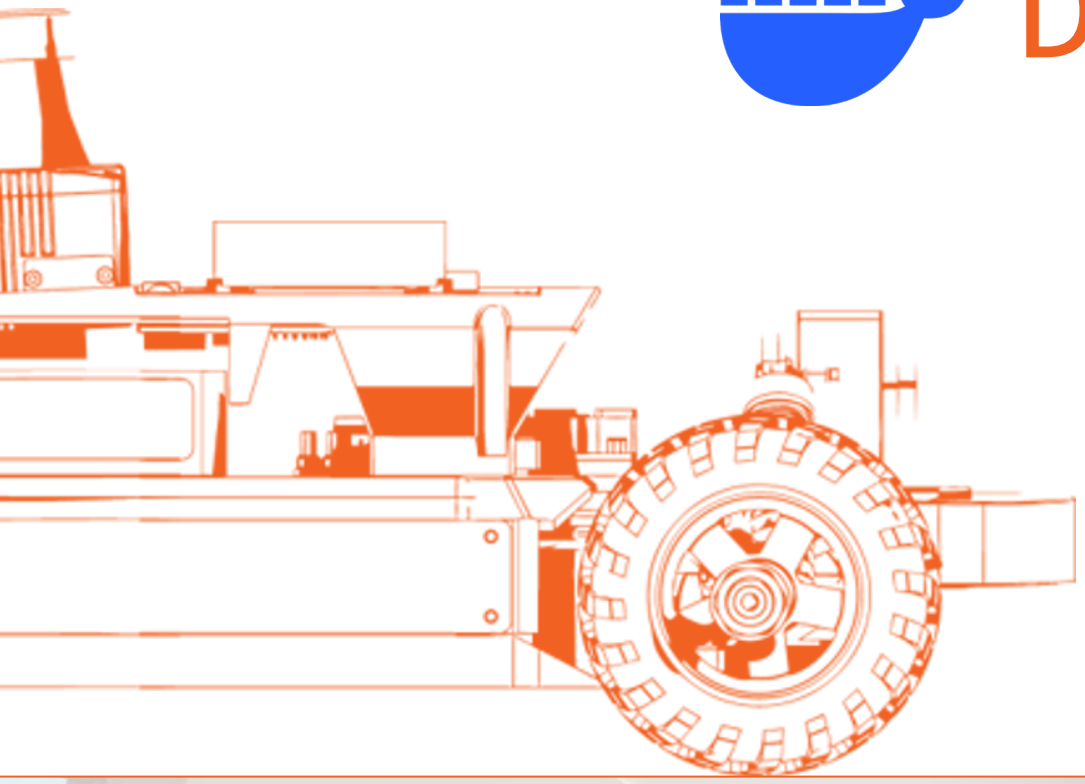
Git 101



- The **working tree** is what you currently see and edit
- The **staging area** is the exact set of changes selected for the next commit
- A **commit** is a saved snapshot with an author, timestamp, identifier, and message
- A **remote repository** is a shared copy of the repository on a hosting platform; GitHub is one example



Docker 101



Docker-101.md

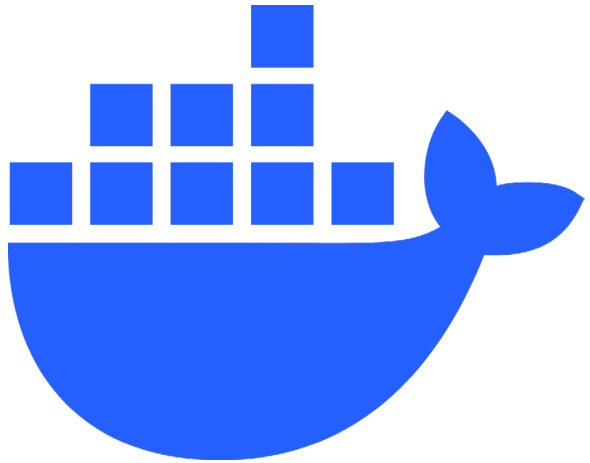
Docker 101

Docker makes the environment portable!

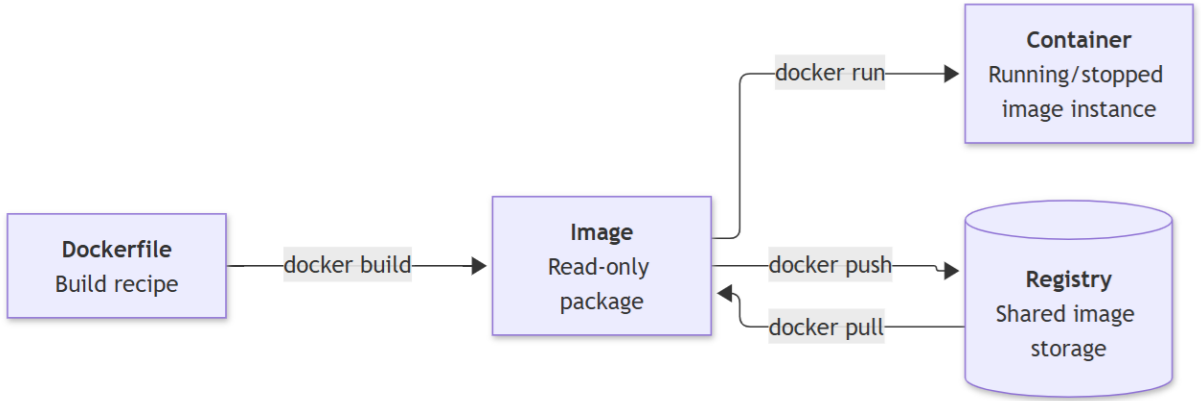
Course context: Docker can give every student a consistent environment for the simulator, robotics middleware, build dependencies, etc. even when host computers differ.

Learning goals:

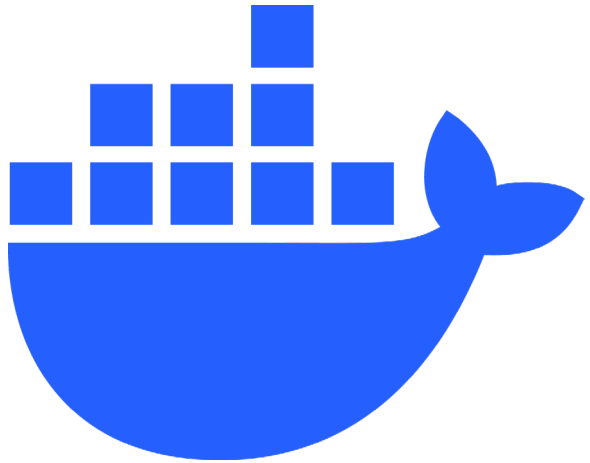
- Explain images, containers, registries, volumes, and bind mounts
- Run and inspect containers
- Map ports and files between host and container
- Write a basic Dockerfile
- Build and tag an image
- Use Docker Compose for a small multi-container setup
- Clean up resources safely



Docker 101



- A **Dockerfile** is a recipe for building an image
- An **image** is a read-only packaged filesystem and configuration
- A **container** is a running or stopped instance of an image
- A **registry** stores images; Docker Hub is one example
- A **volume** stores data managed by Docker
- A **bind mount** exposes a specific host path inside a container



Thank You!

Open to questions and discussion…

